

Quality Assessment of ICA Algorithms for ECG Signal Analysis

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Abstract- The immense scope in the field of biomedical-signal processing Independent Component Analysis (ICA) is gaining momentum due to huge data base requirement for quality testing. The diagnosis of patient is based on visual observation of recorded ECG signals may not be accurate. Many attempts have been carried out to remove noise or artifacts such as interference (few Hz), electromagnetic emission, muscle activities and others from ECG signal. To achieve better understanding, ICA algorithms helps in analyzing ECG signals. This paper describes some algorithms of ICA in brief, such as Fast-ICA, Kernel-ICA, MS-ICA, JADE, EGLD-ICA, Robust ICA etc. The quality & performance of some of the ICA algorithms are tested and analysis of each can be done with respect to Noise/Artifacts, SIR(Signal Interference Ratio), PI(performance Index). In the conclusion tries giving selection type of ICA algorithm for different ECG database.

IndexTerms - ECG(Electrocardiograph), Algorithms, PI, Signal processing, ICA(Independent Component Analysis).

I INTRODUCTION

The detection of electrical properties of the heart can be valuable clinical tool. The ECG is major tool used in clinical situations. In recorded ECGs many types of noise and artifact are present. Early work on noise and artifact reduction in the ECG used either temporal or spatial averaging techniques. Apart, many methods to filter out the noise and artifacts from ECG are only partially successful. On the other hand the filters often lead to a reduction in the amplitudes of the components waves, Q-R-S wave. (As shown in fig 1.) Hence filters to remove interference when it is same frequency range as the cardiac signal. ECG appears to satisfy some of the conditions for ICA: 1) mixing happens linearly at electrodes 2) Time delays in signal transmission are negligible: 3) There appears to be fewer sources than mixtures 4) sources have non-Gaussian voltage distributions. Some violations can happen

just because of movements of the heart such as contraction of the chambers.

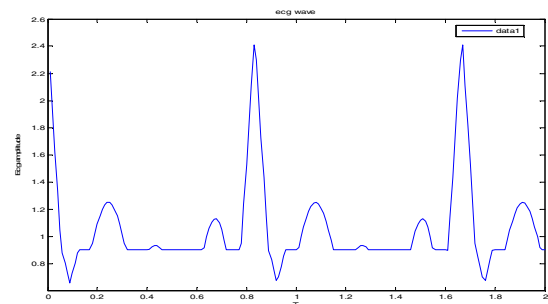


Fig.1 Typical ECG wave

The ECG signal is a very weak time varying signal (about 10 microvolt) and has a frequency between 0.5Hz to 100Hz. The waveforms thus recorded have been standardized in terms of amplitude and phase relationships and any deviation from this would reflect the presence of an abnormality. Abnormal patterns of ECG may be due to undesirable artifacts. Normally ECG is contaminated by power line interference of 60Hz. So it is desired to eliminate this noise and to find how best the signal can be improved.

II. FUNDAMENTALS FOR ICA ALGORITHMS

ICA which was originally proposed to solve the blind source separation problem can be effectively used for ECG analysis. Assume that $x_1, x_2, x_3, \dots, x_n$ be n linear mixtures of n independent signals $s_1, s_2, s_3, \dots, s_n$.

In vector-matrix notation, can be written as follows

$$X = A * S \quad (1.1)$$

where X denotes a column vector having random variables $x_1, x_2, x_3, \dots, x_n$ as its elements, S denotes a column vector having random variables $s_1, s_2, s_3, \dots, s_k$ as its elements, and

A denotes the mixing matrix with elements a_{ij} .
The processing of ICA consists of preprocessing estimating the mixed matrix and computing sources.

DATA PROCESSING STEPS :

Centering: It is nothing but observed data to be used with ICA should have zero mean.

Whitening: Whitening means to transform the observed vector X such that its components become uncorrelated and have unit variances. This is also called sphering. The preprocessing makes the data of ICA estimation simpler and better organized. The main measures of non-gaussianity include- Kurtosis, Negentropy.

III. ICA Algorithms & Results:

A] *Fastica*

It belongs to family of fix point algorithms of ICA, which is based on the iteration to search for maximum of non-gaussianity of variables.

B] *JADE*-

Joint Approximate Diagonalization of Eignmatrices(JADE) is based on higher order cumulant tensors ,of input vector.

C] *Extended Infomax*

It is based on the principle of maximum likelihood estimation . It can detect both super Gaussian and sub Gaussian components.(as shown in fig.3)

D] *Kernel ICA*

It is class of ICA algorithms that deals with projecting the input data in higher dimensions using kernels and then applying optimization methods to find independent components.

E] *MS IC*-

It uses Molgedey and Schuster decorrelation algorithm, having squaring mixing matrix.

F] *EGLD ICA*

It takes skewness of the distribution in practice. EGLD covering whole space of the third and fourth moments fits for more broad of distributions.

IV. PARAMETERS FOR FINDING PERFORMANCE OF ALGORITHMS:

I. Signal to Noise Ratio (SNR)

SNR is an engineering term for the power ratio between a signal (meaningful information) and the background noise. ECG signals normally have a wide dynamic range. SNRs are usually expressed in terms of the logarithmic decibel scale. In decibels, the SNR is 20 times the base-10

logarithm of the amplitude ratio, or 10 times the logarithm of the power ratio:

$$SNR=10 \log (E_s/E_n)$$

where E_s is a average signal amplitude and E_n is average noise amplitude measured within the system bandwidth

Comparison :

SNR : With normal FILTERING techniques
SNR= 23 dB (normal filter)

FASTICA : SNR= 20dB (worst SNR)

Algorithm for calculating SNR (for filtering method)

The following algorithm has been used for calculating SNR:

Step1. Obtain the Original signal and the Filtered (Output) Signal.

Step2. Find the length of the Signal. Obtain the Mean Power of both the Signals.

Step3. Find FFT of the 2 sequences. Correlate the 2 sequences.

Step4. Find Overlap.

Step5. Calculate Signal to Noise Ratio. Convert to Decibel.

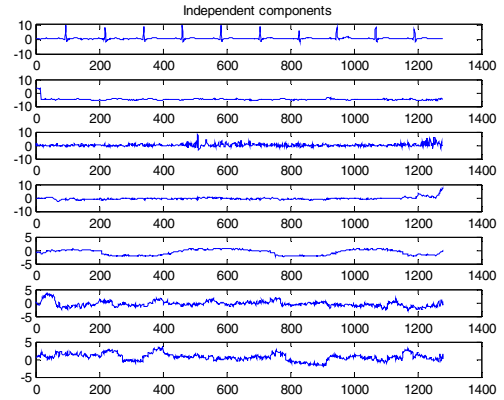


Fig.3 Different component out of signal

II. Performance Index (PI):

The performance index gives performance aspect related to database.

$$PI = \sum_{i=1}^n \left\{ \left(\sum_{k=1}^n \frac{|g_{ik}|^2}{\max_j |g_{ij}|} - 1 \right) + \left(\sum_{k=1}^n \frac{|g_{ik}|^2}{\max_j |g_{ji}|} - 1 \right) \right\}$$

Where

g_{ij} is an ij -th element of the global matrix G

$G=W*A_0$ where

W calculated demixing matrix.

III. Psi

One of the form of judging the performance is Psi

$$\text{Psi} = \{ \text{sum}(\text{sum}(G.^2)) / \text{sum}(\text{max}(G.^2)) - 1 \}$$

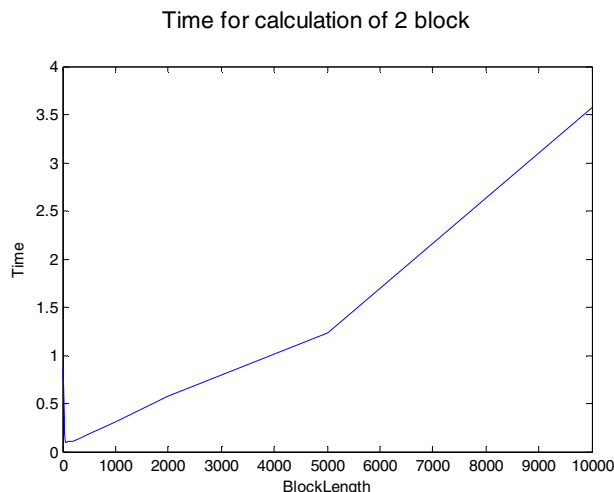


Fig.3 Time for calculation using ICA (FASTICA)

Performance is indicative of the that number of iterations & as well eign values are needed for conclusions.

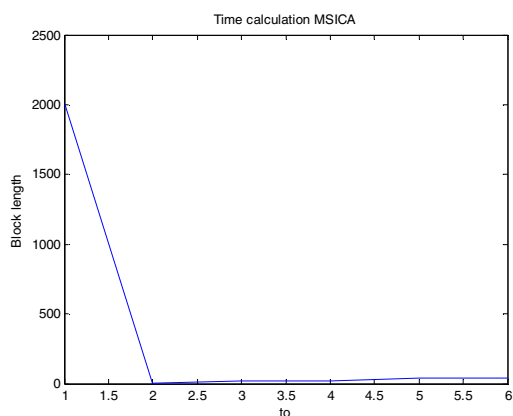


Fig.4. Time calculation for MS ICA for N=1

V. CONCLUSIONS:-

I. Different Algorithm has different performance. When sources have same (lambda) distribution the algorithm of EGLD-ICA

has good results. When sources have same(lambda) distribution the algorithm of EGLD-ICA has good results.

II. FAST-ICA is much more time consuming for longer blocks but less than MS-ICA.

III. FAST-ICA overall performance is good for fewer numbers of blocks (preferably less than 2000 samples).

IV. SNR is much better than normal way of filtering.

V. KICA worked better for more datasets than any other ICA algorithm.

VI. JADE does not depend on gradient optimization techniques, neither on choice of un mixing matrix. This makes it more attractive over others.

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